Immigrants and the Making of America: A Comment*

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Abstract

Sequeira, Nunn and Qian (2020) study the long-run impact of immigration during the Age of Mass Migration on US county-level economic outcomes in 2000 and report large positive effects. They instrument historical immigration with the interaction between county railroad connectivity and European migration to the US, averaged over seven decades (1860–1920). Because this instrument predicts larger inflows in earlier-connected counties, the authors aim to control for this by controlling for *log(years since railroad connection in 2000)*. However, to accommodate the 4% of counties that never received railroads, they instead control for *log(years since railroad connection in 2000 + 1)*, which mainly captures differences between ever- and never-connected counties, rather than between earlier- and later-connected ones. Effectively controlling for the timing of railroad connection substantially reduces the residual variation in the instrument and yields a negative 2SLS estimate, consistent with upward bias in the original estimates: early-connected counties performed better in the long run for reasons unrelated to migration. Because residual variation then depends on county age, the exclusion restriction may still fail; additionally controlling for county age yields an irrelevant first stage. The instrument thus provides insufficient identifying variation to credibly estimate the long-run effects of migration.

1 Introduction

Identifying the long-run effects of historical shocks is challenging. A possible research design isolates historical shocks, aggregates them and examines cross-sectional differences in outcomes in a later time period. A recent and prominent example is Sequeira, Nunn and Qian (2020) (hereafter: SNQ). They study how migration to US counties during the Age of Mass Migration (1860-1920) shapes modern-day (2000) economic outcomes. SNQ instrument for the county-level migration share using the interaction between a binary indicator for railroad connectivity and the total immigration share to the US, averaged across all decades the county existed between 1860 and 1920. They report large effects: a five-percentage-point increase in the average migrant share raises per capita income in 2000 by 13%. They also find that residents of counties with higher historical migration attain more years of schooling.

In this Comment, I show that these findings do not hold once the timing of connection to the railroad network is effectively controlled for. SNQ's instrument varies at the level of the decade of county establishment by the decade of railroad connection: it predicts more migration to counties connected earlier, which are plausibly more prosperous in the long run for reasons unrelated to migration, threatening the exclusion restriction. SNQ acknowledge this and introduce a control for the log number of years since railroad connection in 2000 (hereafter: years of rail).[1] However, SNQ's implementation of this control variable differs. As 127 (4%) counties in the sample never received railways, the authors assign a value of 0 for those counties and use the log(years of rail+1) transformation instead of log(years of rail). This control largely captures differences between ever- and never-connected counties rather than between earlier and later connected counties.

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[1] SNQ write (p. 384): *"One concern with our identification strategy is that the interaction of connection to the railway network and aggregate immigrant inflows might be correlated with how early a county became connected to the railway. To address this, we always control for a measure of how early the county became connected to the railway."*

If counties connected earlier to railroads are systematically more prosperous today, the exclusion restriction is violated and the estimates are upward biased.

I re-analyze SNQ's main result on the effect of migration on the log income per capita in 2000,[2] additionally controlling for a dummy for never connected counties. Together with the log(years of rail+1) control this effectively controls for the timing of railroad connection with a specific functional form. The first stage F-statistic decreases from 12.09 to 4.71 and the weak instrument-robust 90% Anderson-Rubin confidence intervals widen considerably. The 2SLS estimate and Anderson-Rubin confidence intervals turn negative. The sign flip is consistent with an upward bias in the initial estimates: earlier connected counties have mechanically higher instrument values, received more migrants and performed better in the long run for reasons unrelated to migration. However, including log(years of rail + 1) and a never-connected dummy controls for connection timing only parametrically, so residual variation in the instrument remains across counties connected in different decades. Controlling flexibly for the decade of railroad connection through fixed effects yields similar conclusions. Moreover, the instrument also depends systematically on county age, which persists after inclusion of decade of railroad connection fixed effects. Hence, without the strong identifying assumption that county age affects long-run prosperity only through migration the updated estimates do not establish the causal effect of migration on long-run economic performance. Additionally controlling for county age through decade of establishment fixed effects absorbs more variation in the instrument and yields an irrelevant first stage, indicating that the instrument contains insufficient exogenous variation.

The remainder of the paper proceeds as follows. Section 2 discusses SNQ's empirical approach, Section 3 examines the control for time since railroad connection as implemented by SNQ and discusses residual variation in the instrument. Section 4 revisits SNQ's results by (flexibly) controlling for the timing of railroad connection and county age and Section 5 concludes.

2 Empirical approach

SNQ aim to study the causal effect of immigration between 1860 and 1920 on county-level outcomes in the year 2000. As county-level immigration is likely endogenously determined, SNQ use an instrumental variables approach that uses the interaction between whether a county i has railroads and total US-wide immigrant inflows from Europe in the decade before census year t, aggregated over the seven decades between 1860 and 1920. The rationale behind the instrument is that counties that were connected prior to a large influx of immigrants are expected to receive more migrants. SNQ estimate the following two-stage least squares (2SLS) procedure, where Equation 1 is the first stage and Equation 2 is the second stage:[3]

$$\text{Avg Immigrant Share}_{i,s} = \zeta_s + \mu \text{Avg} \widehat{\text{Immigrant}} \text{ Share}_{i,s} + \omega \text{RR Duration}_{i,s} + X_{i,s}\Omega + \epsilon_{i,s} \tag{1}$$

$$Y_{i,s} = \xi_s + \psi \text{Avg Immigrant Share}_{i,s} + \pi \text{RR Duration}_{i,s} + X_{i,s}\Pi + \nu_{i,s} \tag{2}$$

where i indexes a county in state s. Avg Immigrant Share$_{i,s}$ is the average migration share during 1860-1920. The instrument for the immigrant share is given by:

$$\text{Avg} \widehat{\text{Immigrant}} \text{ Share}_{i,s} = \frac{1}{T_i} \sum_{t=T_i^0+10}^{1920} \text{Immigrant Flow}_{t-10} \times I_{i,t-10}^{\text{RR Access}} \tag{3}$$

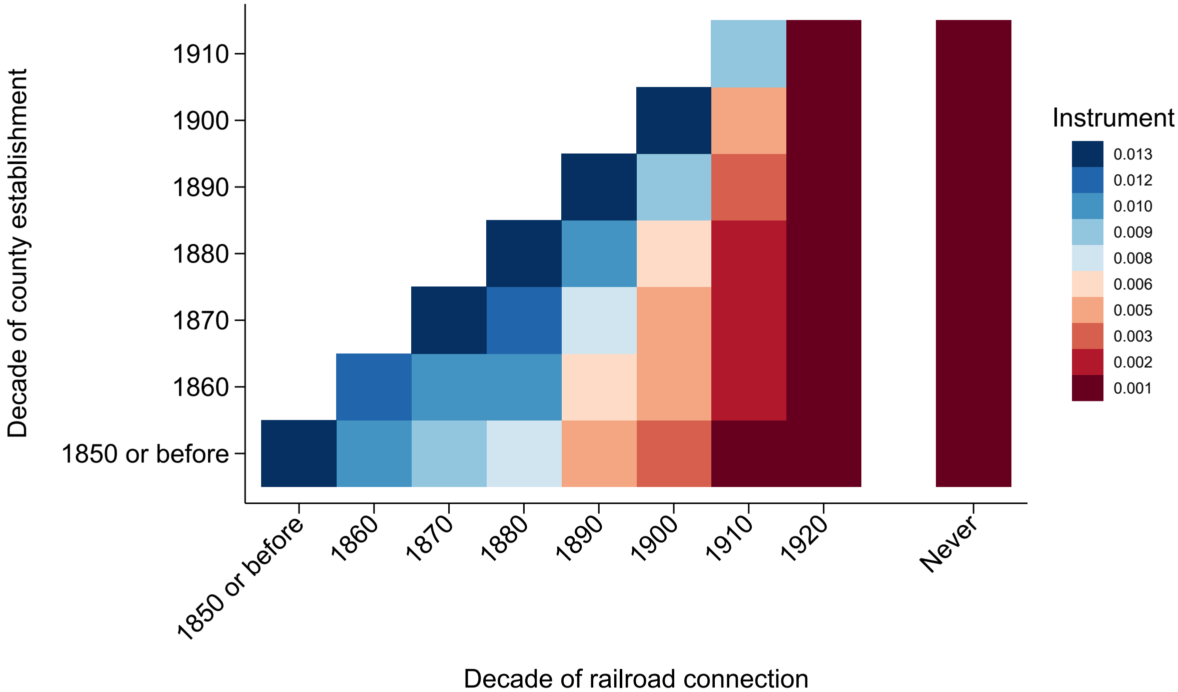
The notation here slightly differs from that in SNQ, to make it explicit that the period is a decade and that the panel is unbalanced.[4] T_i^0 denotes the first decade county i appears in the panel used to construct the instrument, which I refer to as the decade of establishment. $T_i = (1920 - T_i^0)/10$ denotes the number

[2] In the Appendix I show that conclusions are very similar for SNQ's other outcome variables, as those are strongly correlated with log income per capita.

[3] I follow SNQ's notation, in which the instrument is denoted Avg $\widehat{\text{Immigrant}}$ Share$_{i,s}$. Despite the hat, this denotes the instrument itself rather than the fitted value of the endogenous regressor from the first stage, for which SNQ introduce no separate notation in Equation 2.

[4] SNQ write (p. 391): *Since some counties were still in the process of being formed during this period, our panel is unbalanced with counties entering over time. In 1860, there are 1,532 counties in our sample, there are 1,922 counties in 1870; 2,137 in 1880; 2,416 in 1890; 2,692 in 1900; 2,752 in 1910; and 2,935 in 1920.* Counties established during the 1920s or later are not part of the sample.

Figure 1: Instrument value as a function of establishment and railroad cohorts



Notes: Instrument value by decade of establishment and decade of railroad connection. N = 2,935.

of decades the county existed during the Age of Mass Migration. Immigrant Flow_{t-10} is the ratio between total decadal migration to the US between $t-10$ to $t-1$ and the total population of the US in census year $t-10$. $I^{\text{RR Access}}_{i,t-10}$ is a binary indicator for whether a county had railroad access at $t-10$.

The value of the instrument is uniquely determined by the decade of county establishment and the decade of railroad connection: I plot instrument values across both types of decades in Figure 1. For two counties established in the same decade, earlier railroad connection implies a higher instrument value because $I^{\text{RR Access}}_{i,t-10} = 1$ for more periods.[5] Due to pre-existing differences or better economic potential, early connection is plausibly associated with better long-run economic outcomes. This motivates SNQ to control for the timing of railroad connection, RR $\text{Duration}_{i,s}$, which they implement as *log(years of rail+1)*, discussed in detail in the next section. Similarly, for two counties connected in the same decade but established in different decades, the older county has a lower instrument value because T_i is larger. The instrument is therefore also mechanically correlated with county age, which may also be related to long-run economic outcomes—for example, if older counties are located in areas better positioned to benefit from industrialization. SNQ do not acknowledge or discuss this threat to identification, nor do they control for county age.

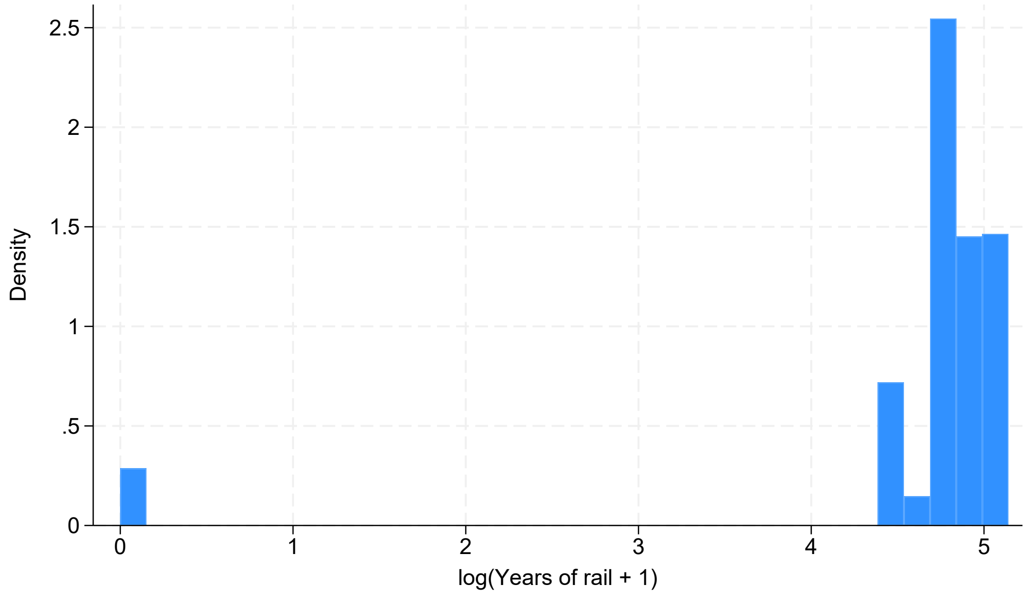
If variation in the instrument driven by county age is undesirable, the instrument could instead be constructed using a balanced panel (i.e. setting $T_i = T = 7$ in Equation 3).[6] In that case, the instrument would vary at the level of decade of railroad connection only and it would be infeasible to control for decade of railroad connection fixed effects. An alternative way of eliminating this dependency is to control flexibly for the decade of county establishment.

SNQ introduce several additional controls. They include state fixed effects (ζ_s and ξ_s) and flexible controls for latitude and longitude to capture geography and historical factors in $X_{i,s}$. In addition, they include two control functions that mirror the structure of the instrument in Equation 3: Immigrant Flow_{t-10} is replaced by (i) the log of decadal US-wide industrialization and (ii) decadal US-wide GDP growth. These variables are included to capture differences in long-run economic performance of counties who had a railroad connection during periods of greater industrialization or strong economic growth. SNQ report Conley (1999) standard

[5] In SNQ's data, railroad connection is an absorbing state: once connected, a county remains connected.

[6] This is the case for SNQ's two alternative instruments, which are discussed in Appendix C.

Figure 2: Histogram of *log(years of rail+1)*



Notes: Histogram of the *log(years of rail+1)* control. N = 2,935.

errors using a five-degree bandwidth (about 400 to 550 km in the US) to account for spatial correlation.

3 Controlling for the timing of railroad connection

3.1 SNQ's implementation

SNQ intend to control parametrically for the timing of connection to the railroad network (p. 392): "$RR\_Duration_{is}$ *is the log number of years, as of 2000, that a county has been connected to the railroad network. The variable is included to address the possibility that our instrument may be correlated with early connection to the railroad network, which could have an independent long-run effect on our outcomes of interest.*" SNQ do not state a rationale for this functional-form choice, nor do they discuss the threat to identification if it is misspecified.

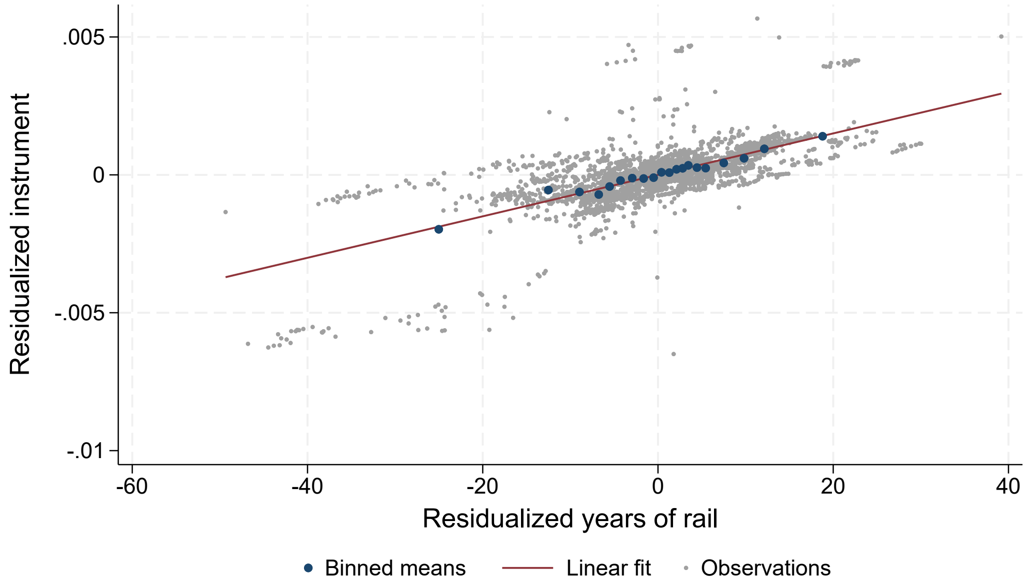
However, the authors instead assign 127 (out of 2,935, 4%) never-connected counties (those without a railroad connection by 1930) a value of 0 and apply the $\log(yearsofrail + 1)$ transformation instead of $\log(yearsofrail)$. This avoids dropping never-connected counties, but in doing so SNQ treat *whether* a county was ever connected as qualitatively equivalent to *when* it was connected.[7] Figure 2 shows the distribution of the *log(years of rail+1)* control. All 127 counties that never received a railroad connection are at 0 and all other counties are in a narrow range between 4.4 and 5.1. The correlation coefficient between *log(years of rail+1)* and a binary indicator for the 127 never-connected counties is $\rho = -0.985$. This shows that the control predominantly captures differences between counties that never had a railroad connection and those that did (the extensive margin), rather than controlling for *when* a county got connected to the railroads, as the authors intended to do (the intensive margin).

3.2 Residual correlation between SNQ's instrument and railroad timing

To examine whether after accounting for all controls the instrument is still correlated with the intensive margin of railroad connection, I residualize (i) the instrument and (ii) the number of years since railroad connection in 2000 using the full set of controls included in Equation 2, including the *log(years of rail+1)*

[7] SNQ only mention the presence of never-connected counties when introducing two additional instruments in Section 7.2, but do not discuss it when introducing the control.

Figure 3: (Binned) scatterplot of the residualized instrument and years since railroad connection



Notes: Scatterplot and binned scatterplot ($N_{bin} = 20$) with trendline of the residualized instrument and the residualized number of years since railroad connection. The control variables used for residualization are the same as in Column 1 of Table 4 of SNQ. N = 2,935. The p-value of the coefficient of a linear regression of the residualized instrument on the residualized years since railroad connection (with an intercept) is $p < 0.001$.

control and industrialization and GDP growth control functions.[8] Figure 3 presents the residual correlation in a binned scatterplot: the residualized instrument is strongly correlated with residualized years since railroad connection.[9] If early- and late-connected counties followed different long-run economic trajectories not due to migration, the exclusion restriction is likely violated. Including *log(years of rail+1)* and an indicator for never-connected counties effectively controls for both the extensive margin and the intensive margin of railroad connection with a specific functional form; I implement this specification in Section 4.

Appendix Figures A3 and A4 also show that substantial, non-linear variation in the instrument across railroad and establishment cohorts remains even after controlling for both controls. Residual variation in the instrument is particularly pronounced for the 1910 and 1920 railroad cohorts and the 1900 and 1910 establishment cohorts. Hence, if there are differences in long-run economic performance between those cohorts that are unrelated to migration, parametrically controlling for the timing of railroad connection may not suffice to fulfill the exclusion restriction.

4 Re-analysis of SNQ's main result

I implement various specifications to effectively control for the timing of railroad connection and county age in Table 1. I focus on SNQ's main outcome—log income per capita in 2000 (Column 1 of SNQ Table 4)—and show results for other outcomes in the Appendix. As the first-stage F-statistic becomes considerably weaker in several specifications, I report 90% Anderson–Rubin confidence intervals (AR CIs), which remain valid for weak instruments (Andrews, Stock and Sun, 2019). The bottom row of Table 1 shows the R^2 of a regression

[8]The control functions for industrialization and GDP growth could, in principle, capture the timing of railroad connection. However, the decadal shocks underlying these controls are only weakly correlated with the decadal migration shocks (see Appendix Figure A1) and thus do not account for the instrument's dependence on railroad timing. Appendix Figure A2a plots the average instrument and control values by decade of railroad connection. The average instrument is lower for counties connected in later decades, whereas the two control functions exhibit a rather flat trend until 1910. The instrument and control function values are 0 for the 1920 and never-connected cohorts. As a result, these controls mainly distinguish the 1920 cohort from earlier connected cohorts, but do not capture gradual differences across counties connected between 1830 and 1910.

[9]SNQ propose two alternative instruments to address concerns related to the spurious correlation with railroad timing. Appendix C shows that these instruments also strongly correlate with the time since railroad connection.

of the instrument on all control variables included in the respective model to assess the residual variation in the instrument.

Column 1 of Table 1 replicates SNQ's baseline specification, which includes all baseline controls including the *log(years of rail+1)* control.[10] Column 2 replaces this control with an indicator for counties that never received railroad access. Reduced form and 2SLS results are nearly unchanged, which is unsurprising since in isolation both controls mainly capture the extensive margin of railroad connection. The coefficient on the never connected dummy indicates that counties that never received a railroad connection have on average similar income levels as those that did receive railroads.

Table 1: Re-analysis of SNQ Table 4, Column 1

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First stage	Average immigrant share, 1860–1920						
Instrument	4.559***	4.512***	3.929**	3.953**	6.463***	3.053	5.587***
	(1.311)	(1.290)	(1.810)	(1.753)	(2.544)	(2.535)	(1.914)
Panel B: Reduced Form	Log income per capita, 2000						
Instrument	11.942***	12.254***	-7.767*	-8.042*	-18.248**	-17.045**	7.414
	(3.629)	(3.626)	(4.474)	(4.584)	(5.813)	(6.226)	(5.696)
Panel C: 2SLS	Log income per capita, 2000						
Average Immigrant	2.619***	2.716***	-1.977	-2.034*	-2.823**		1.327
Share 1860-1920	(1.022)	(1.046)	(1.452)	(1.435)	(1.348)		(1.147)
RI p-value	{0.791}	{0.767}	{0.436}	{0.420}	{0.284}		{0.763}
90% Anderson-Rubin CI	[1.26,5.28]	[1.35,5.43]	[-9.09,-0.12]	[-8.28,-0.15]	[-8.1,-1.23]	$[-\infty,\infty]$	[-0.33,4.44]
log(Years of rail + 1)	0.005		0.348***				
	(0.006)		(0.081)				
Never connected		-0.013	1.571***	0.266***			
		(0.030)	(0.369)	(0.073)			
Years of rail				0.003***			
				(0.001)			
Observations	2935	2935	2935	2935	2935	2935	2935
First-stage F-statistic	12.09	12.23	4.71	5.09	6.46	1.45	8.52
R^2 of IV on controls	0.931	0.930	0.963	0.963	0.980	0.986	0.970
Decade of railroad connection FE					✓	✓	
Decade of county establishment FE						✓	
Sum of shares (μ_i)							✓

Notes: First stage, reduced form and 2SLS estimates based on Column 1 of Table 4 of SNQ. Column 1 presents the original estimates and the additional columns introduce different sets of control variables, indicated by included coefficients in Panel C or checkmarks in the last rows. Because the first stage in Column 6 is very weak and Anderson-Rubin confidence intervals are unbounded, the 2SLS point estimate and randomization-inference p-value are omitted. Coefficients on additional controls shown in Panel C are omitted for brevity in Panels A and B. Conley standard errors using a five-degree window are shown in parentheses. Anderson-Rubin confidence intervals based on Conley standard errors (square brackets) and Randomization Inference p-values (curly brackets) are reported for the 2SLS coefficient of interest. The Anderson-Rubin confidence interval is constructed by inverting the Conley-based Anderson-Rubin test over a grid of 1,001 candidate coefficients from -15 to $+15$ (in steps of 0.03); the reported bounds are the smallest and largest candidate values not rejected at the 10% level, and are set to $\pm\infty$ when ± 15 is not rejected. Kleibergen-Paap F-statistics based on Conley standard errors and the R^2 of a regression of the instrument on all control variables are reported in the last rows of the table. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column 3 includes both controls simultaneously. The *log(years of rail+1)* and never-connected dummy jointly effectively capture the timing of railroad connection with a particular functional form. The additional control captures 46% of the residual variation in Column 1, leaving limited variation.[11] The first stage becomes considerably weaker and the width of the 90% Anderson-Rubin confidence interval doubles.[12] The

[10] I was unable to reproduce the F-statistics reported by SNQ. The F-statistic obtained through Conley (1999) standard errors in Column 1 (12.09) is considerably lower than reported by SNQ (21.222).

[11] The declining residual variation may raise concerns about multicollinearity between the instrument, the industrialization and GDP growth control functions, and the railroad timing controls. Appendix Table A1 reports analogous results excluding the industrialization and GDP growth control functions. Although less variation in the instrument is absorbed, the first stage becomes irrelevant once controlling for the timing of railroad connection.

[12] Appendix Table A8 shows that the first stages of the two alternative instruments that SNQ propose become very weak once both the intensive and extensive margins of railroad access are controlled for; the alternative instruments thus do not

2SLS estimates reverse sign and turn statistically insignificant.[13] Although the 2SLS estimate needs to be interpreted with caution because of the weak first-stage, the drop in the estimate is corroborated by the 90% Anderson-Rubin confidence interval, which only covers negative values.[14] The coefficient on the *log(years of rail+1)* control shows that counties with 10% longer rail access have about 3.3% higher per capita income in 2000, indicating that the timing of railroad connection—not migration—drives the positive estimate in Column 1. Column 4 replaces *log(years of rail+1)* with a linear control for the timing of railroad connection, which yields similar conclusions: the first stage is weak, the 2SLS coefficient is negative and marginally significant, while receiving rail access 10 years earlier is associated with roughly 3% higher income.

The set of controls in Column 3 and 4 may imperfectly capture the confounding relation between the instrument and long-run economic outcomes. Hence, Column 5 includes saturated controls for the timing of railroad through the inclusion of decade of railroad connection fixed effects, including a level for never-connected counties. 71% of residual variation in the instrument is absorbed, the first stage F-statistic drops considerably, and the 2SLS coefficient and AR CI flip sign. The sign flip across Columns 3-5 is consistent with the initial estimates being upward biased as SNQ's specification does not control for the timing of railroad connection: the instrument assigns higher values to earlier connected counties, which are likely more prosperous today for reasons unrelated to migration. However, the residual instrument after (non-)parametrically controlling for the timing of railroad connection is higher for older counties (see Appendix Figure A4). Hence, the estimates in Columns 3-5 do not represent the causal impact of immigration on long-run economic performance, unless one is willing to make strong additional assumptions. Column 6 controls flexibly for county age, absorbing most of the residual variation in the instrument and yielding an irrelevant first stage.[15] Hence, there is insufficient variation that does not depend on the timing of railroad connection or county age.

An alternative identification strategy is offered by the recent shift-share literature, which postdates SNQ. In Appendix A I show that SNQ's instrument has a shift-share structure and argue that the "exogenous shifts" framework of Borusyak, Hull and Jaravel (2022) aligns with SNQ's identifying assumptions: exogenous decadal immigration shocks paired with endogenous shares. Identification can be achieved by controlling for the sum of shares across decades, μ_i, which captures the systematic exposure to the endogenous shares. Column 7 of Table 1 re-estimates SNQ's baseline specification without the *log(years of rail+1)* regressor, controlling for μ_i. The first stage becomes weaker and the 2SLS estimate becomes considerably smaller. In addition, SNQ's instrument assigns treatment to counties in a clustered way which Conley standard errors may not fully account for (see Appendix B for details). Borusyak and Hull (2023) suggest a randomization inference procedure which simulates the decadal migration shocks, recalculates the instrument and calculates a test statistic. Comparing the original to the distribution of simulated test statistics yields p-values that exceed conventional levels in all Columns of Table 1; in other words, randomly reassigned migration shocks would frequently have produced reduced-form estimates at least as extreme as the realized ones. This further supports the conclusion that the instrument contains insufficient exogenous variation.

5 Conclusion

This comment shows that SNQ's instrument controls only for whether a county was ever connected to the railroad, not for when it was connected. Once connection timing is controlled for, the first stage weakens considerably and 2SLS estimates flip sign. The intuition behind this is that earlier-connected counties both

provide sufficient identifying variation.

[13] Appendix Table A5 reports results excluding the 127 counties that never received railroad access. Column 1 uses the baseline specification of column 1 of Table 1, but in this restricted sample *log(years of rail+1)* no longer takes zero values and parametrically controls for the intensive margin of railroad connection; the reduced form and 2SLS estimates closely resemble those in Column 3 of Table 1. Column 2 omits the *log(years of rail+1)* control, which produces results similar to column 1 of Table 1. This confirms that the weaker first stage and sign reversal of the 2SLS coefficient are not merely driven by the exclusion of never-connected counties, but by accounting for the intensive margin of time since railroad connection.

[14] The results for the remaining outcomes in SNQ's Table 4 also turn insignificant or flip sign once including the binary indicator for never-connected counties, as shown in Appendix Table A2. This is unsurprising, since the other outcomes are strongly correlated with income per capita, with absolute correlation coefficients ranging from 0.44 to 0.77. Appendix Table A3 demonstrates that the null results for social outcomes reported in SNQ Table 5 are unaffected. Appendix Table A4 shows that the positive effects on manufacturing output between 1860-1920 and in 1930 become weaker and lose statistical significance once both the extensive and intensive margins of railroad connection are controlled for.

[15] Appendix D discusses why controlling for county age fixed effects renders the first stage irrelevant.

(i) received more migrants during the Age of Mass Migration and (ii) enjoyed better long-run outcomes through channels unrelated to migration. However, the residual variation in turn depends on county age, which arguably violates the exclusion restriction, and conditioning on it leaves the first stage irrelevant. SNQ's instrument therefore lacks sufficient exogenous variation.

How migration during the Age of Mass Migration affected long-run prosperity in the US therefore remains an open empirical question. Future research must rely on stronger sources of exogenous variation in historical settlement patterns. Recent contributions may offer viable and sufficiently powerful alternatives to the railroad-based instrument (Burchardi, Chaney and Hassan, 2019; Obolensky, Tabellini and Taylor, 2024). Another promising direction is to exploit the geography of migrant entry points together with the network structure of railroads to capture variation in travel costs and predict settlement patterns across counties. This is left for further research.

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Online Appendix

(For online publication only)

A Shift-share analysis

In this Appendix, I show that SNQ's instrument has a shift-share structure and can be analyzed using recently developed tools in the literature using slightly different identification assumptions. The shift-share framework I adopt has two ex-ante advantages compared to SNQ's framework. First, it is explicit on how to control for endogeneity in the instrument and with which functional form. Hence, the researcher does not need to select the controls and associated functional form. In the case of SNQ, it is not necessary to control (non-parametrically) for timing of railroad connection and county age, but only for the *sum of shares*. Second, it explicitly accounts for the clustered nature of treatment assignment through the recommended inferential approaches (see Appendix B).

A.1 SNQ's instrument as a shift–share IV

The instrument in Equation 3 can be explicitly written as a shift–share IV:

$$\text{Avg } \widehat{\text{Immigrant}} \text{ Share}_{i,s} = \sum_{t=1860}^{1920} \text{Immigrant Flow}_{t-10} \times RR_{i,t-10} \tag{A1}$$

where the shifts Immigrant Flow_{t-10} are the seven decadal migration shocks. The shares are given by:

$$RR_{i,t-10} = \begin{cases} \dfrac{I_{i,t-10}^{\text{RR Access}}}{T_i}, & \text{if } t-10 \geq T_i^0, \\ 0, & \text{otherwise,} \end{cases} \tag{A2}$$

A feature specific to this setting is that the shock dimension is time. Unlike common shift-share applications, where the shocks are indexed by industry (as in the China shock) or country of origin (as in the migration literature), here they are the seven decadal migration shocks, so serial correlation across shocks is a potential concern for inference (see Appendix B).

A.2 SNQ's instrument in the light of the recent shift-share literature

The recent literature on shift–share instruments offers a framework to clarify identification and inference (Borusyak, Hull and Jaravel, 2022; Goldsmith-Pinkham, Sorkin and Swift, 2020; Adao, Kolesár and Morales, 2019). This literature has shown that there are two pathways to identification with shift–share IVs: assuming that the shifts are exogenous (Borusyak, Hull and Jaravel, 2022), or assuming that the shares are exogenous (Goldsmith-Pinkham, Sorkin and Swift, 2020). As total migration to the US is unlikely to be endogenous to local county-level factors, the former seems plausible. SNQ also provide a compelling argument why the latter is implausible (p. 392): "counties that became connected to the railway network during certain periods ... may have disproportionately benefitted from being connected to the railway".

There are two key identification assumptions underlying the framework of Borusyak, Hull and Jaravel (2022). First, the expected value (but not its dispersion) of the shocks should be the same for all shock dimensions. In the case of SNQ, this implies that the expected value of the migration shocks should be equal across decades. As the shocks are not trending but rather fluctuate from decade to decade, this assumption seems reasonable. This assumption can in principle also be relaxed by the inclusion of additional controls. Second, there should be a sufficient number of uncorrelated shocks. Borusyak, Hull and Jaravel (2025) write: "it [the exogenous shifts approach] requires many shifts $g_i \ldots, g_K$. Otherwise, if K is a small number, the shifts may by chance be correlated with unobservables even if they are truly random". In the case of SNQ, there are only seven decadal shocks, so residual variation in the instrument may be correlated with the timing of railroad connection and county age. This also has consequences for inference, as discussed further in Appendix B.

If those assumptions are satisfied, identification can be achieved by controlling for the sum of shares across the shock dimension. If the sum of shares differs across observations, units with a larger sum of shares are systematically more exposed to the shocks and have more extreme instrument values in expectation. If the shares are endogenous, this correlation between the instrument and the sum of shares implies a violation of the exclusion restriction. In the case of SNQ, the sum of shares across decades can be written as:

$$\mu_i = \sum_{t=1860}^{1920} RR_{i,t-10} = \frac{R_i}{T_i} \tag{A3}$$

R_i is the number of decades a county was connected to the railroad between 1860 and 1920, and T_i is the number of decades the county existed during this period. Similar to the instrument itself (Figure 1), the sum of shares is positively correlated with earlier railroad connection and negatively correlated with county age, as shown in Figure A5. Table A7 shows that the sum of shares is strongly correlated with the instrument, the average immigrant share, and log income per capita in 2000, even when baseline controls are included.

A.3 Results

I re-estimate SNQ's baseline specification without the *log(years of rail+1)* control but with the sum of shares, μ_i. Column 7 of Table 1 shows that the first stage based on Conley standard errors remains moderately strong after controlling for μ_i. At the same time, the 2SLS coefficient declines substantially in magnitude and becomes statistically insignificant at the 10% level. As discussed above, even when conditioning on μ_i, the instrument may still correlate with railroad timing or county age. Figures A3 and A4 show that this is indeed the case: both the residualized instrument and residualized income in 2000 are higher for earlier-connected and older counties.

I examine how results change when (additionally) controlling for the timing of railroad connection and county establishment in Table A6. Columns 1-4 report the results of specifications that include linear controls for railroad timing and county age, with and without the sum of shares μ_i. Inclusion of only a linear control for years since railroad connection—capturing the intensive margin only partially—yields estimates that are smaller and marginally significant. Once μ_i is also included, the coefficients are no longer statistically distinguishable from zero. If county age is instead controlled for linearly, the 2SLS estimate turns insignificant. These findings suggest that when controlling parametrically for the sum of shares (and parametrically for railroad timing or county age) the first stage already becomes considerably weaker. In addition, the 2SLS estimates and Anderson-Rubin confidence intervals are mostly insignificant, even though the standard errors used are prone to overstate precision, which I address in the next section.

B Randomization inference

The small number of shifts also complicates inference, as a single decadal shock can exert substantial influence on the instrument values of many counties. Standard asymptotic inference does not account for the complex dependence structure of shift–share designs (Adao, Kolesár and Morales, 2019), in line with the recommendation to cluster standard errors at the level of treatment assignment (Moulton, 1990; Abadie et al., 2023). This weakness is also present in SNQ's design as treatment is assigned in a clustered fashion based on the level of decade of railroad connection $\times$ decade of county establishment. Standard errors should account for this by clustering on the level of treatment assignment (Moulton, 1990; Abadie et al., 2023). SNQ use spatial Conley (1999) standard errors, but those assume independence for units jointly assigned to treatment that are spatially distant.

Borusyak, Hull and Jaravel (2022) show that the 2SLS estimator can be re-expressed as a shift-level IV regression, allowing for valid inference using (cluster-)robust standard errors. In our setting, however, SNQ's instrument is constructed from fewer shocks than the number of included covariates, rendering this approach infeasible. In such cases, Borusyak and Hull (2023) recommend a randomization inference (RI) procedure using the Hodges-Lehmann T-statistic. Because it is a randomization test, this procedure is valid in finite samples and does not rely on the strength of the instrument, so it remains valid under weak identification. Other test statistics can also be calculated, such as the t-statistic of the first stage to assess instrument strength.

The RI procedure repeatedly draws shocks from a plausible counterfactual shock distribution, recomputes the instrument, and calculates the simulated test statistic. To assess how likely the original test statistic is to arise by chance, it is compared to the distribution of simulated test statistics. The first step concerns specifying realistic counterfactual shocks. I specify the following three shock generation processes:

1. Randomly permuting the realized shocks across decades. This is equivalent to re-drawing without replacement. This yields 7! possible permutations.

2. Drawing shocks from the set of realized shocks, with replacement. This yields 7^7 possible permutations.

3. Drawing shocks from a normal distribution associated with the mean (0.076) and standard deviation (0.025) of the realized shocks

I draw $N_b = 999$ instances of these shocks, recalculate the instrument and calculate the following test statistics using the original instrument and each simulated instrument:

- First-stage t-statistic
  To assess how strong the original first stage is compared to the simulated first stages, I calculate the first-stage t-statistic.

- Hodges-Lehmann T-statistic
  Borusyak and Hull (2023) recommend to calculate the Hodges-Lehmann T-statistic to assess the strength of the 2SLS evidence: T= $\frac{1}{N}\sum_i (Z_i - \theta\mu_i)(y_i - b^{cand}x_i)$, where Z_i is the (recalculated) instrument, θ is the mean of the decadal migration shocks, μ_i the sum of shares and b^{cand} is the candidate parameter value. $\theta\mu_i$ is the expected value of the instrument. We are interested in the case $b^{cand} = 0$. Hence, T is equivalent to the covariance between the "recentered" instrument and the outcome. Note that recentering (subtracting the expected value of the instrument $\theta \cdot \mu_i$) yields a secular shift independent of the realization of the shocks and instrument.

- Reduced form coefficient β^{RF}
  To assess how large the realized reduced form coefficient is relative to the coefficient obtained from randomly drawn shocks, I calculate the reduced form coefficient.

- Reduced form t-statistic τ^{RF}
  Compared to the reduced form coefficient β, the t-statistic has the advantage that it is also informative of the strength of the evidence, not solely the size of the coefficient.

After calculation of these test statistics, I compare how extreme the test statistics based on the actual instrument are in comparison to those from the simulated instruments. Specifically, for each test statistic I calculate two-sided Randomization Inference p-values in the following way:

$$\hat{p}^T_{\mathrm{RI}} = \frac{2}{N_b} \min\left(\sum_{n=1}^{N_b} \mathbf{1}\left\{T^{(n)} \geq T^{\mathrm{orig}}\right\}, \sum_{n=1}^{N_b} \mathbf{1}\left\{T^{(n)} \leq T^{\mathrm{orig}}\right\} \right) \tag{A4}$$

Here, $n = \{1, ..., N_b\}$ indexes the simulations, and *orig* indicates that it concerns the original realized instrument. T denotes any of the four test statistics described above. This test can also be used to construct valid confidence intervals through inversion of the test described above for different candidate values b^{cand} on a grid of 1,001 points from -15 to $+15$ (in steps of 0.03). For each b^{cand}, the randomization inference p-value is calculated. If the p-value exceeds α, the candidate b^{cand} falls inside the $(1-\alpha)$ confidence interval. If the p-value is smaller than α, the candidate b^{cand} falls outside the confidence interval.

B.1 Results

For each specification and test statistic, I plot the distribution across permutations and indicate the corresponding test statistic from the original instrument with a dashed vertical line. Figure A6 reports the results for SNQ's baseline specification. The realized 2SLS T-statistic is not extreme relative to its permutation distribution: most randomly drawn decadal migration shocks yield more extreme test statistics.

Randomization inference therefore leads to markedly different conclusions compared with the conventional asymptotic inference based on Conley standard errors used by SNQ: while the conventional p-value is 0.010, the RI p-value is 0.791. The reduced form t-statistics are highly asymmetric and skewed toward positive values. 53% of t-statistics exceed 2, implying that for randomly reassigned shocks, conventional inference frequently rejects the null in favor of a positive effect of the instrument on log income per capita in 2000. The first stage t-statistic is still relatively extreme, suggesting that the first stage is relevant.

Controlling for μ_i, as the shift-share framework of Borusyak, Hull and Jaravel (2022) prescribes, partially mitigates these issues. Figure A7 shows that the test statistic distributions become more symmetric and centered around zero. The first-stage t-statistic is still relatively large (p=0.028), but the T-statistic for the 2SLS estimate is not extreme at all (p=0.763). The reduced form results further illustrate the shortcomings of asymptotic inference for this specification: 61% of simulated t-statistics lie below -2 or above 2. Figures A8 and A9 show that sampling shocks with replacement or from a normal distribution, with the same mean and standard deviation as the realized shocks, leads to similar conclusions.

Tables 1 and A6 report RI p-values based on the Hodges-Lehmann T-statistic across specifications; these are consistently much larger than those based on Conley standard errors. The RI framework also allows for the construction of valid confidence intervals by inverting the test on the Hodges-Lehmann T-statistic for different candidate parameter values. I calculate those confidence intervals using the specification that controls for μ_i, permuting the migration shocks across decades. Figure A10 shows that the 90% confidence interval of the effect of a 1 percentage point higher migration share during the Age of Mass Migration on per capita income in 2000 ranges from a reduction of 12% to an increase of 8%, which is uninformatively large. Hence, the instrument contains insufficient variation to study the question at hand.

C Alternative instruments

SNQ employ several alternative instrumental variables to alleviate remaining endogeneity concerns. The first uses weather shocks to predict decadal migration from Europe. This design reduces endogeneity concerns related to European migration reacting to railroad connection in counties with large growth potential, but does not address the issue outlined in Section 3.1.

SNQ also introduce two alternative instruments in Column 2 and 3 of their Table 11 motivated by concerns about the correlation between the original instrument and timing of railroad connection. When doing so, the authors include the never-connected dummy but remove the *log(years of rail+1)* control in these specifications.[1] Hence, the alternative specification also does not control for time since railroad connection.

The first alternative instrument calculates the average migration shock across decades after receiving a railroad connection. The second alternative instrument only considers migration flows in the decade following the decade of railroad connection. In contrast to the original instrument, both alternative instruments do not depend systematically on county age through division by T_i. Both alternative instruments can be written in the form of Equation A1, with the same shifts but different shares. The share part of the first alternative instrument can be written as:

$$RR^{avg.connected}_{i,t-10} = \begin{cases} \dfrac{1}{N_i^{RR}}, & \text{if } I^{\text{RR Access}}_{i,t-10} = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{A5}$$

SNQ here define N_i^{RR} as the number of *years* a cohort was connected to the railways between 1830 and 1920.[2] Let R_i^0 denote the decade in which a county received railroad connection. The sum of shares for this instrument is given by:

$$\frac{\min(1920 - R_i^0, 70)/10}{(1930 - R_i^0)} \tag{A6}$$

[1] SNQ write (footnote 57): *"Since the predicted average immigrant share instrument for counties that are never connected to the railway network is zero, the specifications include an indicator variable for whether the county was never connected to the railway."* However, this issue is not unique to the alternative instruments, but also arises for the original instrument.

[2] In their paper SNQ write (p. 412): where "N_i^{RR} is the number of time periods". However, in the replication package it is defined as the number of years.

For example, for those first connected to railroads in the 1910s, the sum of shares is $\frac{1}{20}$. For those counties never connected or first connected in the 1920s the sum of shares is equal to 0. This instrument yields a very weak first stage.[3]

The share part of the second alternative instrument can be written as:

$$RR^{first}_{i,t-10} = \begin{cases} 1, & \text{if } I^{\text{RR Access}}_{i,t-10} = 1 \text{ and } I^{\text{RR Access}}_{i,t-20} = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{A7}$$

For counties that received railways access before 1850, the share is 1 for the first decade ($t = 1860$) and 0 for all subsequent decades. The sum of shares is equal to 1 for counties connected in 1910 or earlier and 0 for counties connected in the 1920s or later.

Although those alternative instruments are intended to be uncorrelated with the timing of railroad connection, the shares still depend on it: exposure to shocks is not equal across railroad cohorts, and the realized instrument still depends on the realization of the seven decadal migration shocks. To illustrate that these alternatives also have lower instrument values for later decades, I plot the average instrument values by decade in Figure A2b, analogous to Figure A2a. This shows that the alternative instruments are still lower for later connected counties. Figure A11 shows that both alternative instruments also exhibit a residual correlation with the timing of railroad connection. Hence, the same concern about upward bias also applies to these alternative instruments. Table A8 shows that when including both the *log(years of rail+1)* and the never connected control, the first stages for both instruments turn very weak. This can be explained by the fact that both alternative instruments do not systematically depend on county age, contrary to the original instrument. Appendix D discusses why the dependence on county age is important for the first stage.

D Why does the first stage vanish when controlling for decade of establishment FE?

Table 1 and Table A6 show that the instrument remains relevant when controlling (flexibly) for the timing of railroad connection and when adding a linear control for county age. However, Column 5 of Table A6 introduces saturated controls for the decade of county establishment, which renders the first stage extremely weak; adding decade of railroad connection fixed effects on top (Column 6 of Table 1) leaves it equally uninformative. This shows that the conclusions about whether the instrument has sufficient exogenous variation depend on the assumptions about whether and how county age affects long-run outcomes.

Why do decade of county establishment fixed effects render the first stage irrelevant? Figure A12 shows that the residualized (using SNQ's controls) average immigrant share follows an inverse-U pattern across decades of establishment. Counties established during the early mass migration period (1860–1880) exhibit substantially higher immigrant shares (3–12 percentage points higher), while both older and younger counties display lower shares. The residualized instrument in Figure A4a mirrors this pattern: counties founded between 1860 and 1890 have positive residuals even after accounting for the timing of railroad connection. This inverse-U pattern can be understood using the construction of the instrument: older counties have smaller shares (larger T_i; see Equation A2) and younger counties have shares of 0 for most decades (i.e. they accumulated fewer migration shocks; see Equation A1).

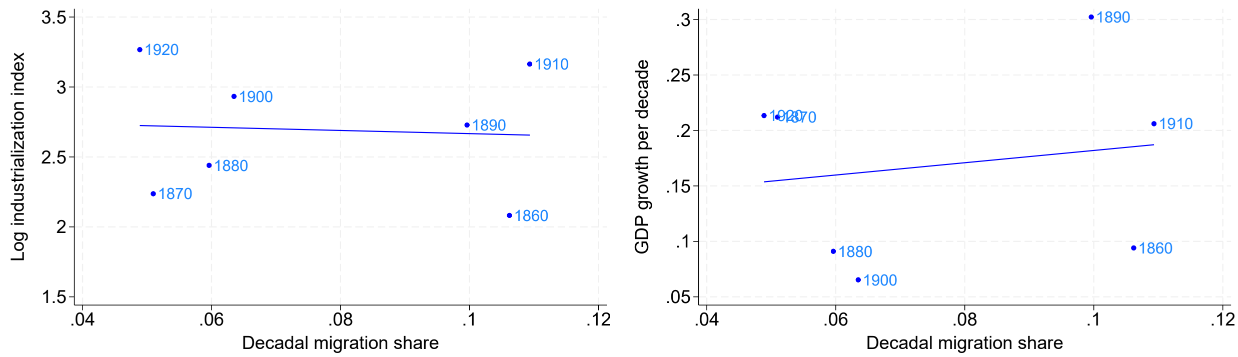
The coinciding variation across establishment cohorts explains the first stage effect, and also explains why linear controls, as introduced in Table A6, do not render the first stage irrelevant. The pattern is plausible. Counties founded at the onset of mass migration may have attracted more immigrants than long-established counties or those not yet formed as labor demand may be particularly high in newly created counties and migrants are more willing to settle there than natives. SNQ do not discuss this and may not have intended to rely on variation in the instrument that depends on county age.

Whether the (residual) correlation with county age is desirable is less clear. Establishment cohort-specific factors may threaten the exclusion restriction. Older (often coastal) counties may still be richer today due to persistent geographic advantages. Conversely, some younger counties may have successfully attracted new industries.

[3] Column 2 of Table 11 of SNQ reports the first-stage F-statistic of 18.803. However, I was unable to reproduce this: the F-statistic based on Conley standard errors with a 5 degree window is only 3.7.

E Additional Figures

Figure A1: Correlation of decadal shocks: industrialization and GDP growth vs. migration

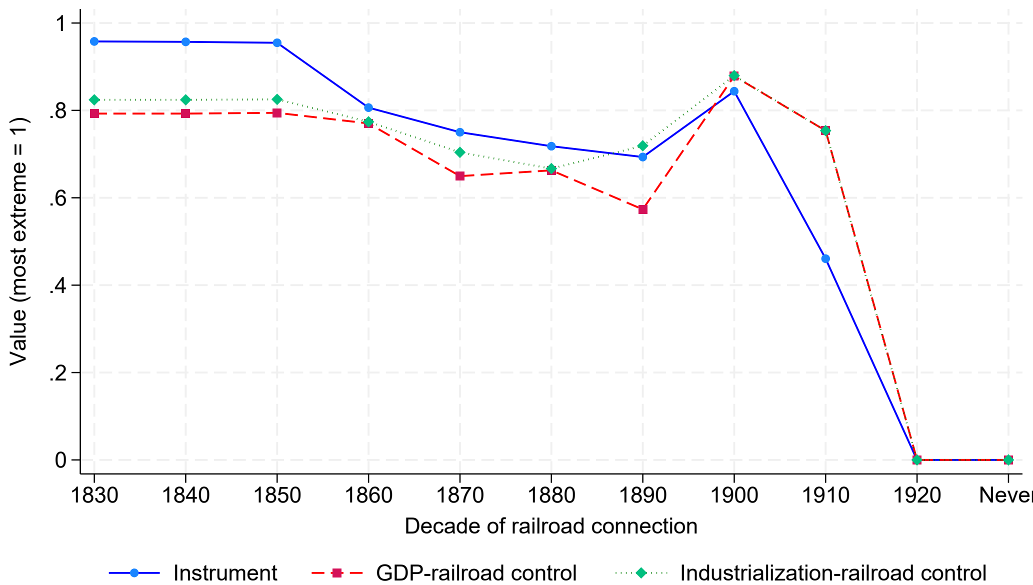


(a) Industrialization

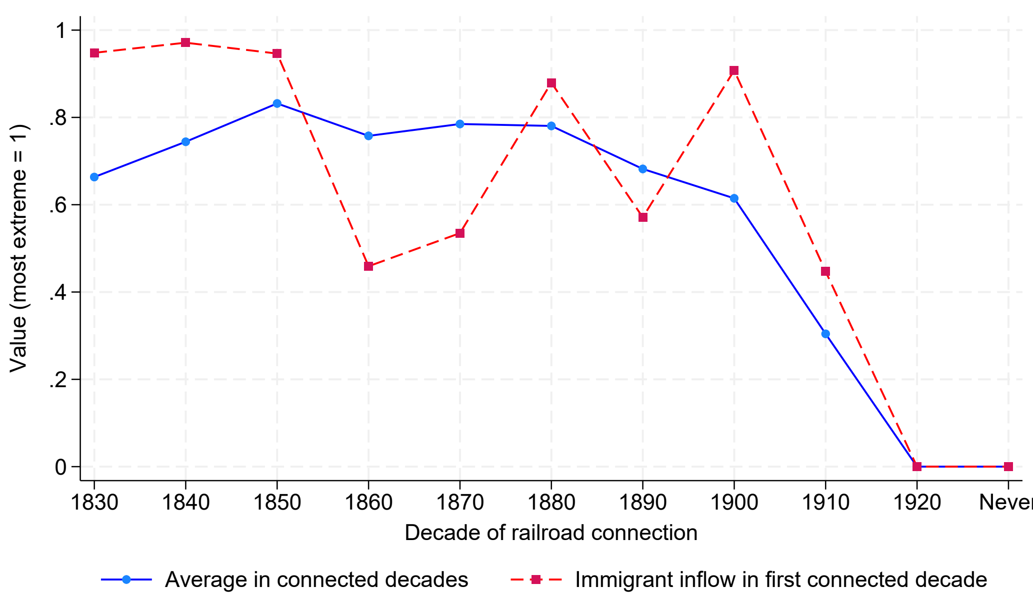
(b) GDP growth

Notes: Each marker indicates one of the seven decades between 1860 and 1920; the line is a linear fit. The horizontal axis is the decadal migration shock, European immigration to the US normalized by the total US population. The vertical axis is the lagged log industrialization index in Panel (a), and lagged GDP growth per decade between $t-10$ and t in Panel (b). The correlation of the seven decadal migration shocks is $\rho = -0.07$ with the log industrialization index and $\rho = +0.17$ with GDP growth per decade.

Figure A2: Average instrument control function values by decade of railroad connection.



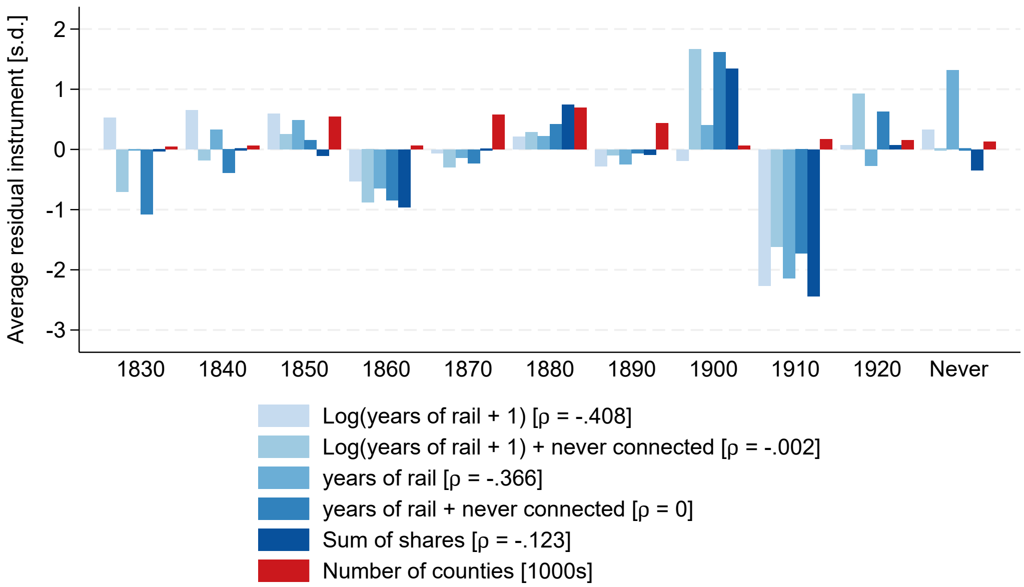
(a) Average instrument and industrialization and GDP growth control functions



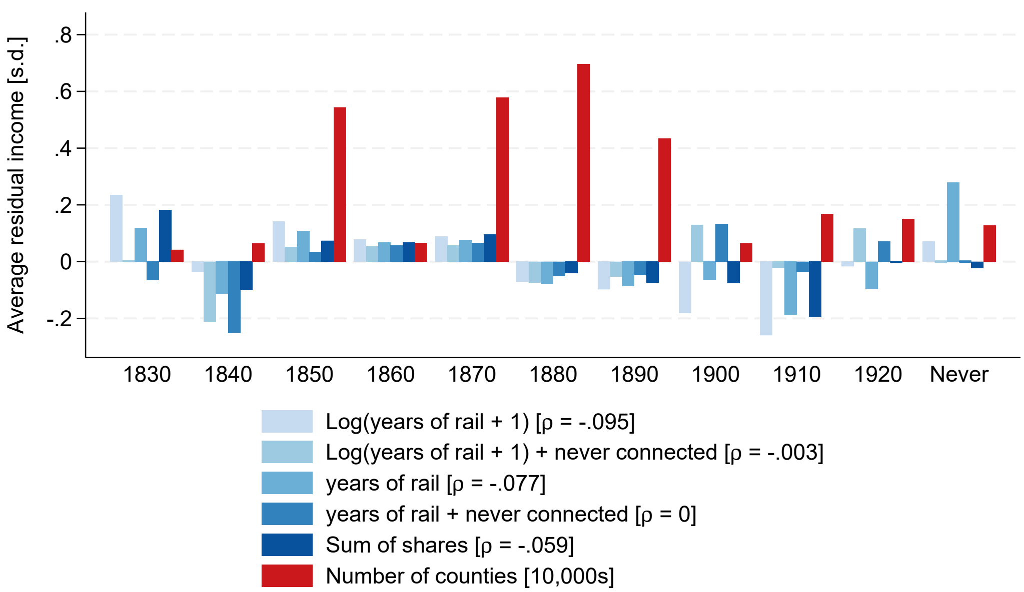
(b) Alternative instruments

Notes: a) Average instrument and industrialization and GDP growth control function values by decade of railroad connection. b) Average alternative instrument values used in SNQ Table 11 by decade of railroad connection. For visualization purposes I divide each time series by its most extreme value across all counties and decades of railroad connection. N = 2,935.

Figure A3: Residualized instrument and residualized log income per capita by decade of railroad connection



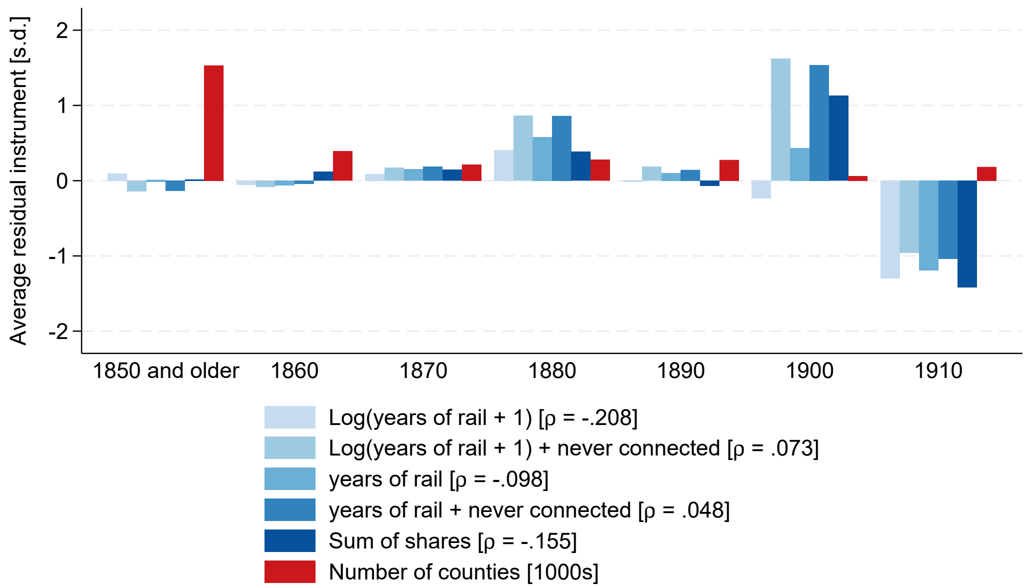
(a) Instrument



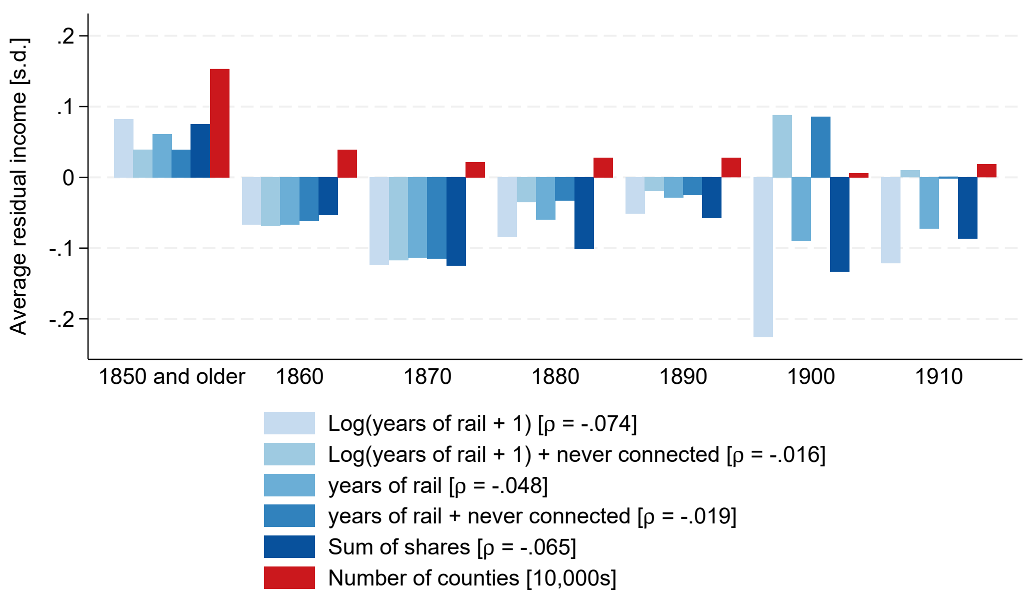
(b) Log income per capita in 2000

Notes: Average residualized and standardized a) instrument and b) log income per capita in 2000 by decade of railroad connection, for different sets of controls. Each specification includes the baseline controls of Table 4 of SNQ (excluding the control for *log(years of rail+1)*, unless noted otherwise) and the controls mentioned in the legend. The legend reports the correlation between the residualized instrument and the decade of railroad connection in square brackets, on the sample of ever-connected counties. The number of observations by cohort is indicated by the red bars.

Figure A4: Residualized instrument and residualized log income per capita in 2000 by decade of establishment



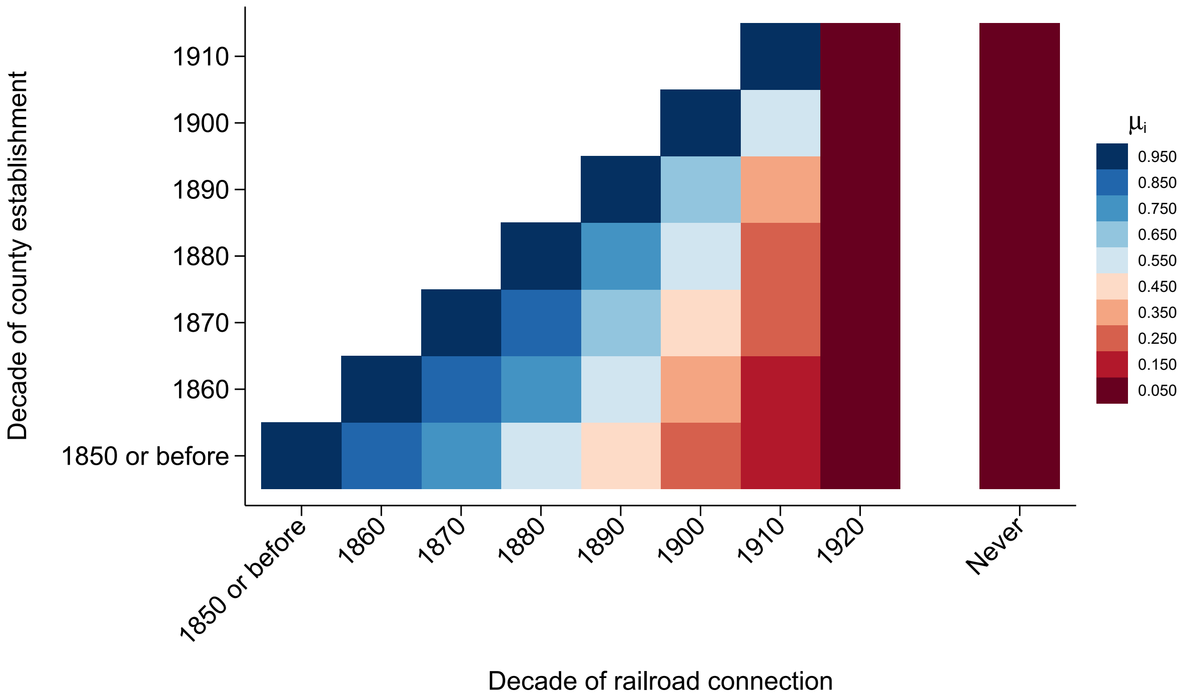
(a) Instrument



(b) Log income per capita in 2000

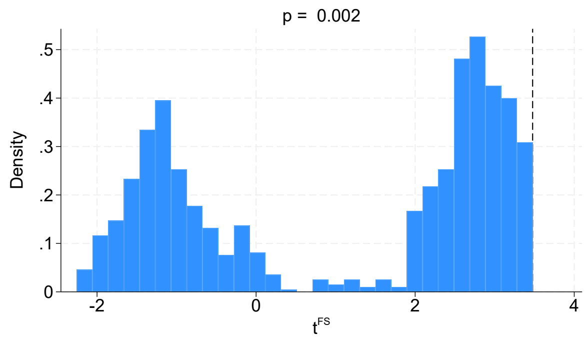
Notes: Average residualized and standardized a) instrument and b) log income per capita in 2000 by decade of establishment, for different sets of controls. Each specification includes the baseline controls of Table 4 of SNQ (excluding the control for *log(years of rail+1)*, unless noted otherwise) and the controls mentioned in the legend. The legend reports the correlation between the residualized instrument and the decade of establishment in square brackets. The number of observations by cohort is indicated by the red bars.

Figure A5: Sum of shares μ_i as a function of establishment and railroad cohorts

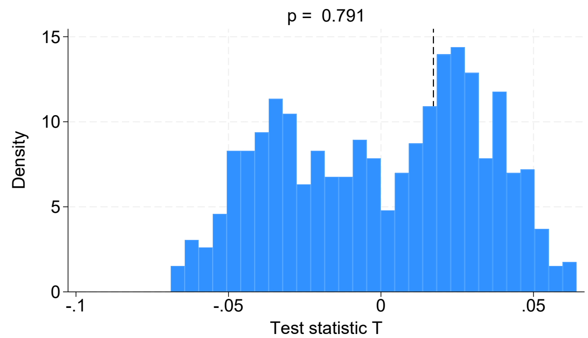


Notes: Sum of shares μ_i by decade of establishment and decade of railroad connection. N = 2,935.

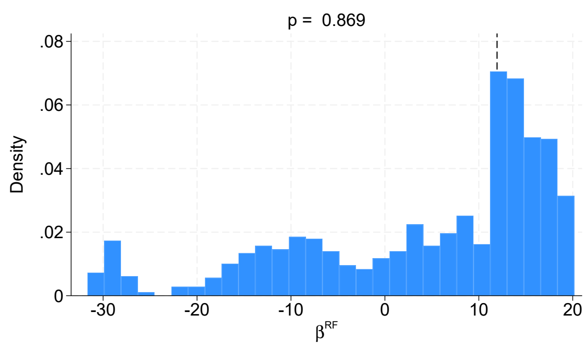
Figure A6: Distribution of RI test statistics by permuting decadal shocks, original specification



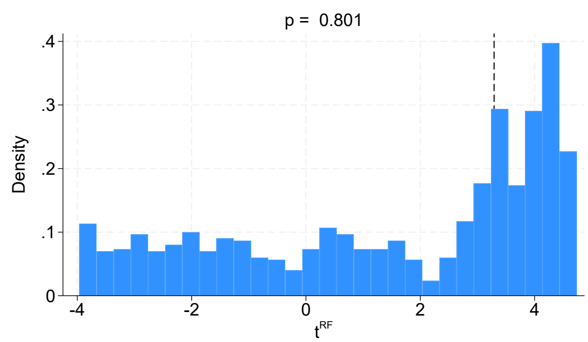
(a) First-stage t-statistic



(b) Hodges-Lehmann T-statistic



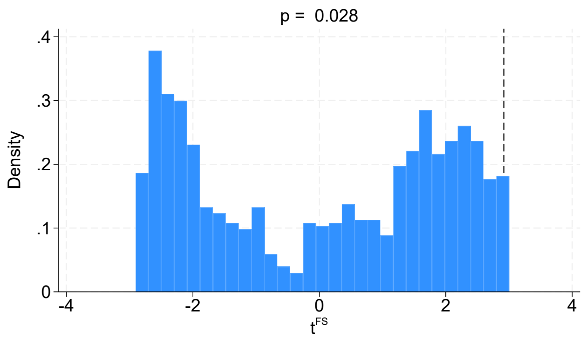
(c) Reduced form coefficient



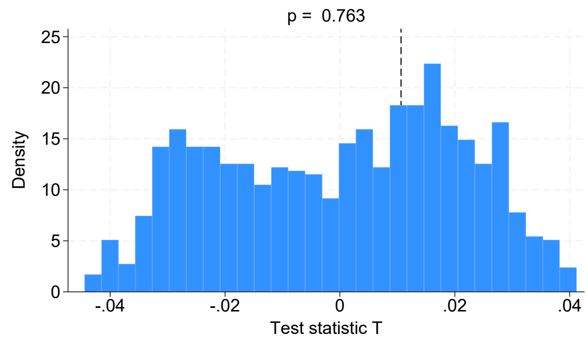
(d) Reduced form t-statistic

Notes: Distribution of Randomization Inference test statistics, based on the specification in Table 4 of SNQ, which includes all baseline controls as well as the *log(years of rail+1)* control. Shocks are drawn from the set of 7! permutations of the seven decadal shocks across the seven decades, without replacement. Panel (a) shows the t-statistic of the coefficient on the instrument in the first stage. Panel (b) shows the distribution of the Hodges-Lehmann statistic T, given by T= $\frac{1}{N}\sum_i (Z_i - \theta\mu_i)(y_i - b^{cand}x_i)$, where θ is the mean of the seven decadal shocks and the null hypothesis is that $b^{cand} = 0$. Panel (c) shows the distribution of coefficient estimates on the instrument from the reduced form regression, and Panel (d) the distribution of the corresponding t-statistics. Dashed lines indicate the value of the respective statistic for the realized shock. t-statistics are calculated using Conley standard errors adjusted for spatial correlation using a five-degree window. Randomization Inference p-values are defined as the share of simulations that yield a statistic more extreme than the realized one, times two. Based on N = 999 randomly drawn permutations.

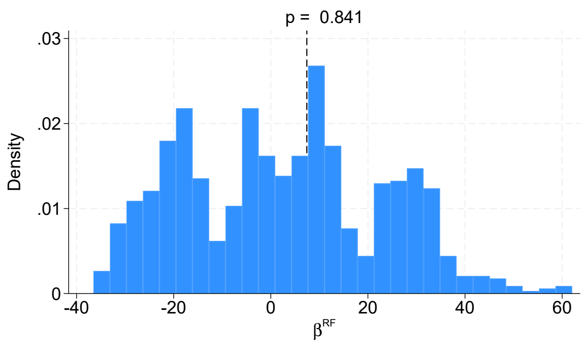
Figure A7: Distribution of RI test statistics by permuting decadal shocks, controlling for μ_i



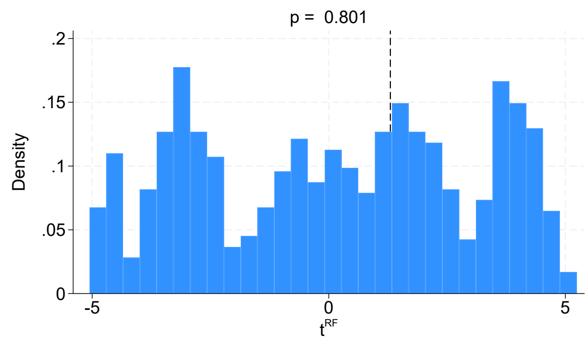
(a) First-stage t-statistic



(b) Hodges-Lehmann T-statistic



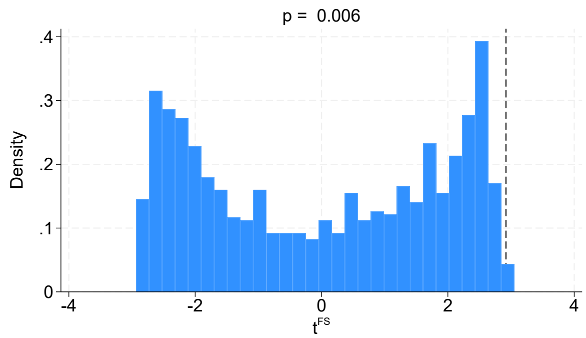
(c) Reduced form coefficient



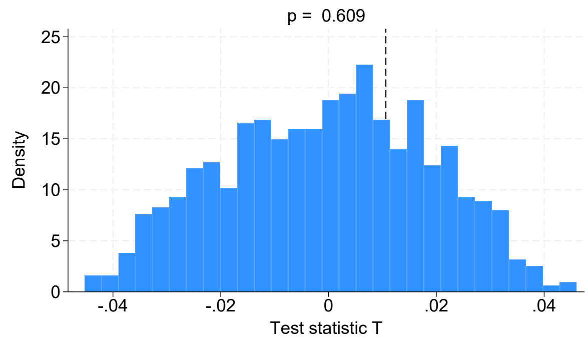
(d) Reduced form t-statistic

Notes: Distribution of Randomization Inference test statistics, based on a specification that includes all baseline controls as well as the sum of shares μ_i, but excludes the control for *log(years of rail+1)*. Shocks are drawn from the set of 7! permutations of the seven decadal shocks across the seven decades, without replacement. For a description of the test statistics shown in Panels (a)–(d) and of the procedure, see the notes to Figure A6. Based on N = 999 randomly drawn permutations.

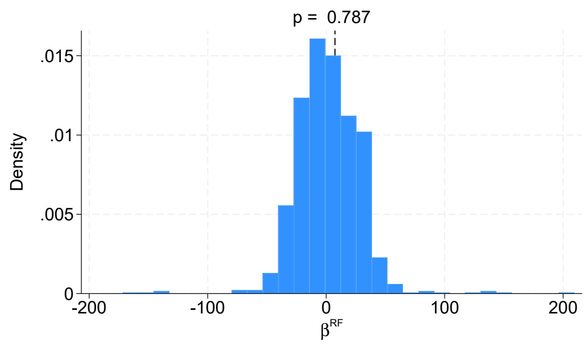
Figure A8: Distribution of RI test statistics by redrawing shocks with replacement, controlling for μ_i



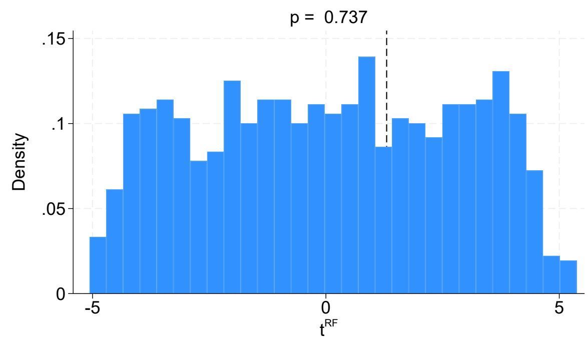
(a) First-stage t-statistic



(b) Hodges-Lehmann T-statistic



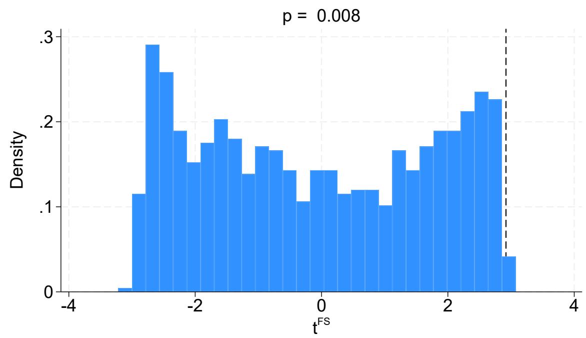
(c) Reduced form coefficient



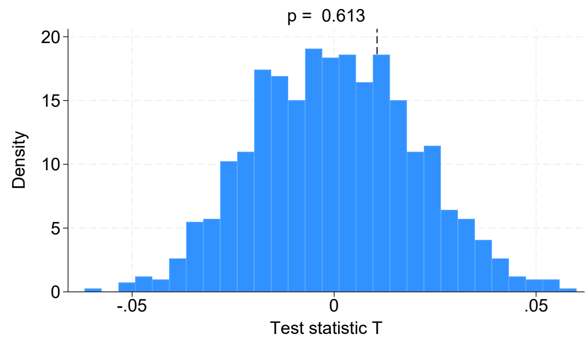
(d) Reduced form t-statistic

Notes: Distribution of Randomization Inference test statistics, based on a specification that includes the sum of shares μ_i but excludes the control for *log(years of rail+1)*. Shocks are drawn from the set of permutations of the seven decadal shocks, with replacement. For details on the procedure, see notes to Figure A6. Based on N = 999 permutations.

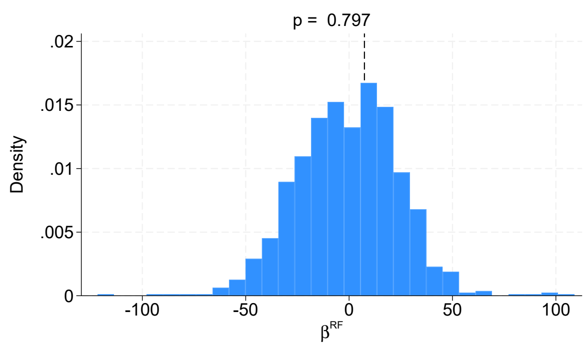
Figure A9: Distribution of RI test statistics by drawing from normal distribution, controlling for μ_i



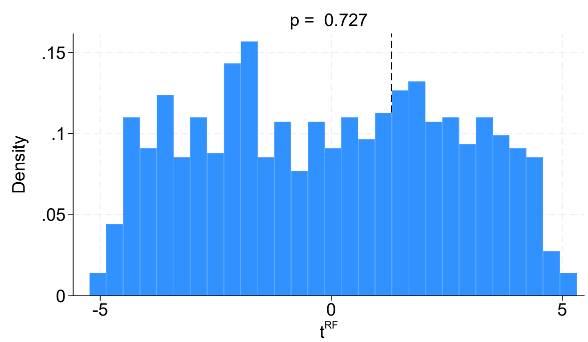
(a) First-stage t-statistic



(b) Hodges-Lehmann T-statistic



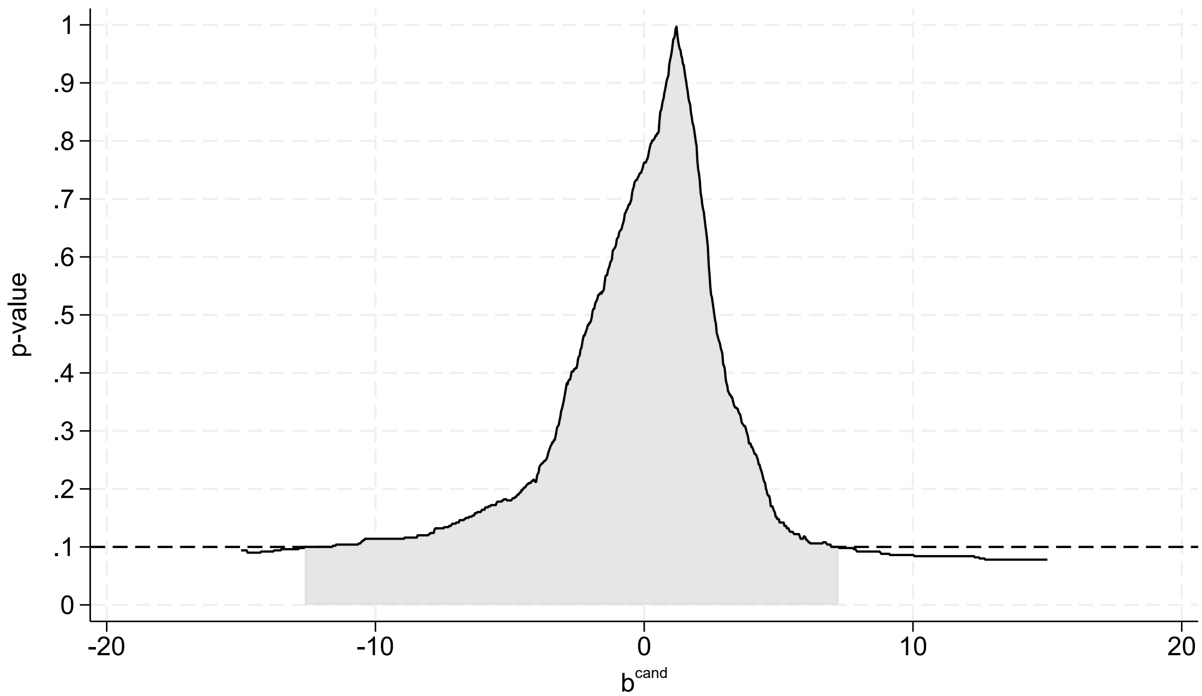
(c) Reduced form coefficient



(d) Reduced form t-statistic

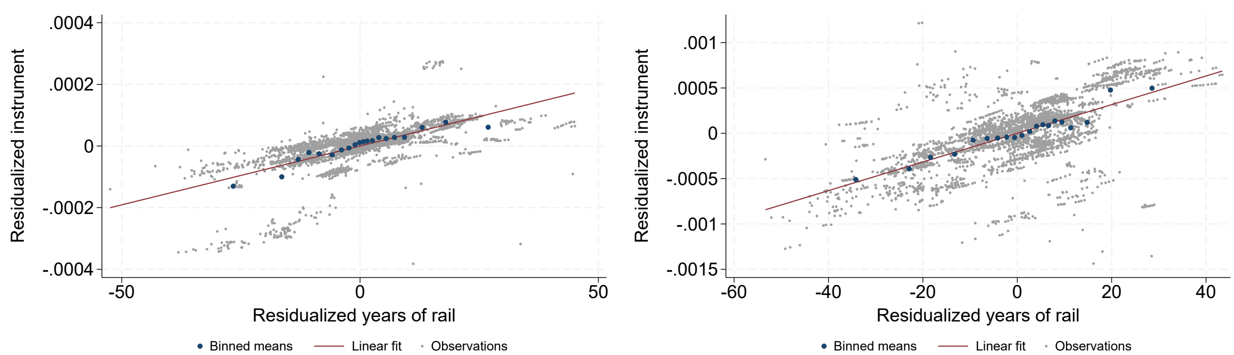
Notes: Distribution of Randomization Inference test statistics, based on a specification that includes the sum of shares μ_i but excludes the control for *log(years of rail+1)*. Shocks are drawn from a normal distribution with a mean (0.076) and standard deviation (0.025) equal to that of the seven realized shocks. For details on the procedure, see notes to Figure A6. Based on N = 999 permutations.

Figure A10: Randomization Inference Confidence Intervals



Notes: Randomization inference p-values for different candidate values b^{cand} based on the Hodges-Lehmann T-statistic as suggested by Borusyak and Hull (2023). The specification includes all baseline controls (excluding the *log(years of rail+1)* control) and the sum of shares μ_i. The instrument is recalculated by randomly permuting the shocks across decades 999 times. The p-value for every b^{cand} is obtained according to the procedure discussed in Appendix B, on a grid of 1,001 candidate values between -15 and $+15$ (in steps of 0.03). The area shaded in gray shows the area where the test does not reject the null using $\alpha = 0.1$. The implied 90% confidence interval ranges from -13.2 to 7.5. The corresponding effect size of a one percentage point higher migration share ranges from -12% to +8%.

Figure A11: Binned scatterplot of the residualized alternative instruments and time since railroad connection

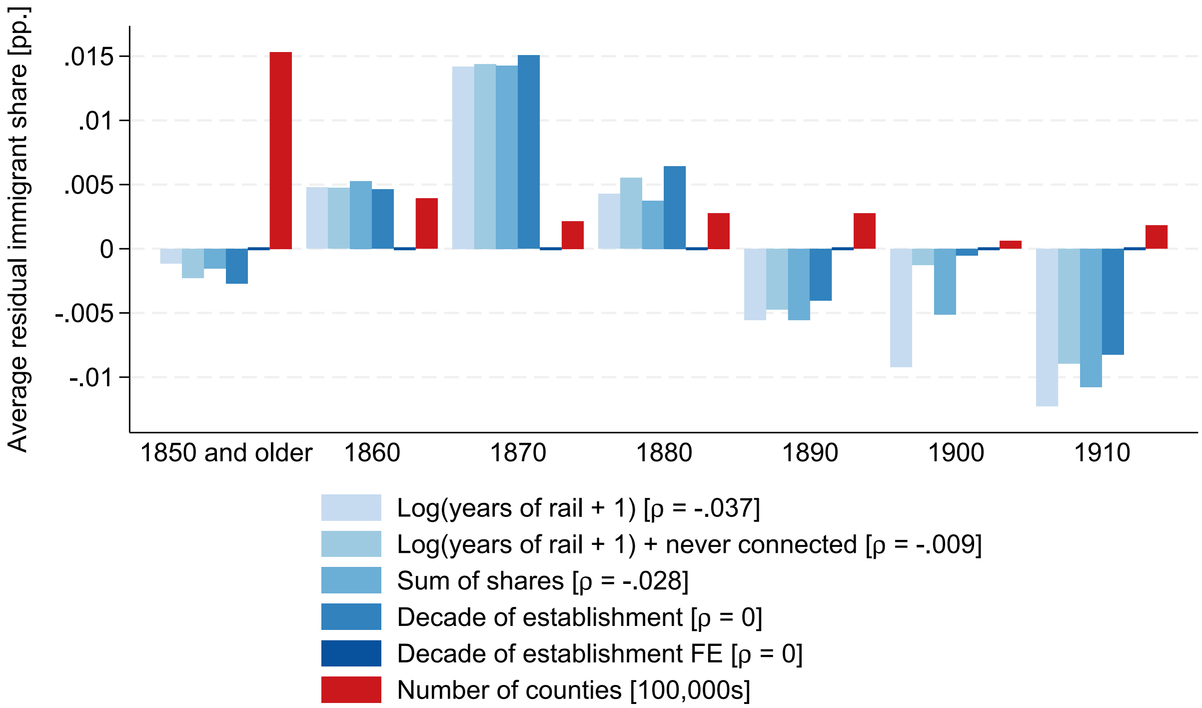


(a) Average over connected decades

(b) First connected decade

Notes: Binned scatterplot ($N_{bin} = 20$) with trendline of the residualized alternative instruments and the residualized number of years since railroad connection. The instruments are (a) the average migration shock over connected decades and (b) the migration shock in the first decade following railroad connection. The control variables used for residualization are the same as Columns 2 and 3 of Table 11 of SNQ, which include a never-connected dummy but do not include the control for *log(years of rail+1)*. N = 2,935. In both panels, the p-value of the coefficient of a linear regression of the residualized instrument on the residualized years since railroad connection (with an intercept) is $p < 0.001$.

Figure A12: Residualized average immigrant share (1860-1920) by decade of county establishment for different controls



Notes: Average residualized immigrant share between 1860 and 1920 by decade of county establishment, for different sets of controls. Each specification includes the baseline controls of Table 4 of SNQ (excluding the control for the *log(years of rail+1)*, unless noted otherwise) and the controls mentioned in the legend. The legend also reports the correlation between the average immigrant share (1860-1920) and the decade of establishment in square brackets. The number of observations by cohort is indicated by the red bars.

F Additional Tables

Table A1: Re-analysis of SNQ Table 4, Column (1), excluding the shift-share-like control functions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First stage	Average immigrant share, 1860–1920						
Instrument	1.590***	1.584***	0.635	0.676	0.770	1.655	6.048***
	(0.479)	(0.458)	(0.844)	(0.775)	(0.999)	(1.836)	(1.835)
Panel B: Reduced Form	Log income per capita, 2000						
Instrument	6.679***	7.146***	-2.799	-2.029	-2.498	2.395	10.531**
	(1.695)	(1.707)	(2.130)	(2.090)	(2.400)	(3.943)	(4.997)
Panel C: 2SLS	Log income per capita, 2000						
Average Immigrant Share 1860-1920	4.201***	4.512***					1.741*
	(1.734)	(1.813)					(1.026)
log(Years of rail + 1)	0.008		0.426**				
	(0.007)		(0.207)				
Never connected		-0.031	1.942**	0.296**			
		(0.035)	(0.960)	(0.134)			
Years of rail				0.003**			
				(0.001)			
Observations	2935	2935	2935	2935	2935	2935	2935
First-stage F-statistic	11.02	11.95	0.57	0.76	0.59	0.81	10.86
R^2 of IV on controls	0.507	0.431	0.796	0.770	0.852	0.931	0.967
Decade of railroad connection FE					✓	✓	
Decade of county establishment FE						✓	
Sum of shares (μ_i)							✓

Notes: First stage, reduced form and 2SLS estimates based on Column 1 of Table 4 of SNQ, excluding the two shift-share-like control functions for industrialization and GDP growth; all other baseline controls of SNQ are included. The columns follow the same specifications as Table 1. Because the first stages are very weak in Columns 3–6, the corresponding 2SLS point estimates are omitted. Coefficients on additional controls shown in Panel C are omitted for brevity in Panels A and B. Conley standard errors using a five-degree window are shown in parentheses. Kleibergen-Paap F-statistics based on Conley standard errors and the R^2 of a regression of the instrument on all control variables are reported in the last rows. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Re-analysis of SNQ Table 4

	Log income per capita in 2000		Share below poverty line		Unemployment rate		Urbanization rate		Average years of schooling	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: OLS										
Average Immigrant Share 1860-1920	0.243***	0.211***	0.015	0.022	0.020*	0.024**	0.949***	0.913***	0.020	-0.080
	(0.130)	(0.130)	(0.028)	(0.028)	(0.015)	(0.015)	(0.184)	(0.189)	(0.307)	(0.306)
Panel B: Reduced Form										
Instrument	11.942***	-7.767*	-2.229***	2.228**	-1.876***	0.010	22.382***	5.655	41.925***	-11.143
	(3.629)	(4.474)	(0.777)	(1.308)	(0.500)	(0.639)	(6.820)	(9.774)	(10.562)	(13.148)
Panel C: 2SLS										
Average Immigrant Share 1860-1920	2.619***	-1.977	-0.489***	0.567*	-0.411***	0.002	4.909***	1.439	9.195***	-2.836
	(1.022)	(1.452)	(0.209)	(0.409)	(0.151)	(0.162)	(2.008)	(2.584)	(3.392)	(3.488)
Observations	2935	2935	2935	2935	2935	2935	2935	2935	2935	2935
First stage F-statistic	12.09	4.71	12.09	4.71	12.09	4.71	12.09	4.71	12.09	4.71
log(years of rail+1)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Never connected		✓		✓		✓		✓		✓

Notes: OLS, reduced form and 2SLS estimates of Table 4 of SNQ. Odd Columns control for all baseline controls including the *log(years of rail+1)* control, even Columns additionally control for a dummy for never connected counties. Conley standard errors using a five-degree window are shown in parentheses. Kleibergen-Paap F-statistics based on Conley standard errors are reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Re-analysis of SNQ Table 5

	Social capital		Voting turnout		Total crime rate		Crimes against persons		Crimes against property	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: OLS										
Average Immigrant Share 1860-1920	-0.048**	-0.047**	-0.071**	-0.073**	0.008***	0.008***	0.002***	0.002***	0.004***	0.004***
	(0.029)	(0.030)	(0.046)	(0.046)	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Panel B: Reduced Form										
Instrument	0.210	1.441	1.244	1.471	0.086	0.055	0.020	0.012	0.054	0.031
	(0.958)	(1.192)	(1.662)	(2.060)	(0.070)	(0.108)	(0.013)	(0.022)	(0.053)	(0.075)
Panel C: 2SLS										
Average Immigrant Share 1860-1920	0.046	0.364	0.271	0.378	0.019	0.014	0.004	0.003	0.012	0.008
	(0.209)	(0.326)	(0.347)	(0.517)	(0.017)	(0.028)	(0.003)	(0.005)	(0.012)	(0.019)
Observations	2934	2934	2925	2925	2935	2935	2935	2935	2935	2935
First stage F-statistic	12.09	4.71	12.09	4.71	12.09	4.71	12.09	4.71	12.09	4.71
log(years of rail+1)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Never connected		✓		✓		✓		✓		✓

Notes: OLS, reduced form and 2SLS estimates of Table 5 of SNQ. Odd Columns control for all baseline controls including the *log(years of rail+1)* control, even Columns additionally control for a dummy for never connected counties. Conley standard errors using a five-degree window are shown in parentheses. Kleibergen-Paap F-statistics based on Conley standard errors are reported.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Re-analysis of SNQ Table 6

	Log Average Manufacturing Output per Capita				Log Average Manufacturing Output per Establishment				Log Number Establishments per 10,000 Inhabitants			
	1860-1920		1930		1860-1920		1930		1860-1920		1930	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: OLS												
Average Immigrant Share 1860-1920	3.476***	3.423***	4.216***	4.094***	3.301***	3.275***	3.343***	3.222***	0.319**	0.263*	0.783***	0.782***
	(0.631)	(0.638)	(0.796)	(0.802)	(0.537)	(0.540)	(0.648)	(0.653)	(0.249)	(0.250)	(0.248)	(0.249)
Panel B: Reduced Form												
Instrument	40.765*	13.672	74.736***	-4.844	20.778	0.042	71.924***	-4.705	32.710***	9.121	2.079	-0.204
	(33.988)	(41.084)	(26.368)	(42.989)	(29.227)	(35.276)	(23.653)	(38.610)	(6.462)	(10.231)	(6.765)	(9.776)
Panel C: 2SLS												
Average Immigrant Share 1860-1920	9.014*	3.487	16.197***	-1.116	4.594	0.011	15.588***	-1.084	7.253***	2.315	0.453	-0.047
	(8.460)	(11.195)	(7.343)	(9.821)	(6.838)	(8.998)	(6.868)	(8.800)	(2.389)	(2.859)	(1.467)	(2.400)
Observations	2805	2805	2463	2463	2805	2805	2463	2463	2804	2804	2462	2462
First-stage F-statistic	6.99	1.97	6.01	2.22	6.99	1.97	6.01	2.22	6.99	1.97	6.01	2.22
log(years of rail+1)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Never connected		✓		✓		✓		✓		✓		✓

Notes: OLS, reduced form and 2SLS estimates of Table 6 of SNQ. Odd Columns control for all baseline controls including the *log(years of rail+1)* control, even Columns additionally control for a dummy for never connected counties. Conley standard errors using a five-degree window are shown in parentheses. Kleibergen-Paap F-statistics based on Conley standard errors are reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Re-analysis of SNQ Table 4, Column 1, restricted to counties ever connected by rail

	(1)	(2)
Panel A: First stage	Average immigrant share, 1860–1920	
Instrument	3.788**	4.524***
	(1.854)	(1.318)
Panel B: Reduced Form	Log income per capita, 2000	
Instrument	-8.775**	12.641***
	(4.357)	(3.596)
Panel C: 2SLS	Log income per capita, 2000	
Average Immigrant	-2.316*	2.794***
Share 1860-1920	(1.639)	(1.063)
RI p-value	{0.404}	{0.797}
90% Anderson-Rubin CI	[-13.02,-0.39]	[1.41,5.64]
log(Years of rail + 1)	0.386***	
	(0.079)	
Observations	2808	2808
First-stage F-statistic	4.17	11.78
R^2 of IV on controls	0.951	0.908

Notes: First stage, reduced form and 2SLS estimates based on Column 1 of Table 4 of SNQ, restricted to the 2,808 counties that were ever connected to the railroad. Column 1 includes all baseline controls including the *log(years of rail+1)* control, which in this restricted sample parametrically captures the intensive margin of railroad connection; Column 2 removes the *log(years of rail+1)* control. The coefficient on the *log(years of rail+1)* control shown in Panel C is omitted for brevity in Panels A and B. Conley standard errors using a five-degree window are shown in parentheses. Anderson-Rubin confidence intervals based on Conley standard errors (square brackets) and Randomization Inference p-values (curly brackets) are reported for the 2SLS coefficient of interest. The Anderson-Rubin confidence interval is constructed by inverting the Conley-based Anderson-Rubin test over a grid of 1,001 candidate coefficients from -15 to $+15$ (in steps of 0.03); the reported bounds are the smallest and largest candidate values not rejected at the 10% level, and are set to $\pm\infty$ when ± 15 is not rejected. Kleibergen-Paap F-statistics based on Conley standard errors and the R^2 of a regression of the instrument on all control variables are reported in the last rows of the table. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Controlling for additional timing-related factors

	(1)	(2)	(3)	(4)	(5)
Panel A: First stage	Average immigrant share, 1860–1920				
Instrument	4.610***	5.653***	3.707**	4.798**	1.391
	(1.465)	(1.923)	(1.460)	(1.977)	(1.358)
Panel B: Reduced Form	Log income per capita, 2000				
Instrument	6.560*	3.243	3.669	-1.627	7.035*
	(3.790)	(5.585)	(3.219)	(5.303)	(4.747)
Panel C: 2SLS	Log income per capita, 2000				
Average Immigrant Share 1860-1920	1.423*	0.574	0.990	-0.339	
	(0.875)	(1.016)	(0.950)	(1.096)	
RI p-value	{0.821}	{0.847}	{0.931}	{0.933}	
Anderson-Rubin 90% CI	[0.09, 3.54]	[-1.14, 2.91]	[-0.45, 3.84]	[-2.76, 2.1]	$[-\infty,\infty]$
Observations	2935	2935	2935	2935	2935
First-stage F-statistic	9.91	8.64	6.45	5.89	1.05
R^2 of IV on controls	0.945	0.972	0.942	0.973	0.958
Years of railroad connection	✓	✓			
Years since county establishment			✓	✓	
Sum of shares (μ_i)		✓		✓	
Decade of county establishment FE					✓

Notes: First stage, reduced form and 2SLS estimates with alternative controls for timing of railroad connection and county age. All models include the baseline controls of Table 4 of SNQ, except for the *log(years of rail+1)* control. Column 1 controls linearly for the years of railroad connection in 2000, which partially captures the intensive and extensive margins of railroad connection. Column 2 adds a control for the sum of shares μ_i. Column 3 controls linearly for the years since county establishment in 2000, Column 4 additionally controls for the sum of shares μ_i. Column 5 includes fixed effects for the decade of county establishment; because the first stage is very weak and the Anderson-Rubin confidence interval is unbounded, the 2SLS point estimate and randomization-inference p-value are omitted. Conley standard errors using a five-degree window are shown in parentheses. Anderson-Rubin confidence intervals (square brackets) and Randomization Inference p-values (curly brackets) are reported for the key 2SLS coefficient of interest. The Anderson-Rubin confidence interval is constructed by inverting the Conley-based Anderson-Rubin test over a grid of 1,001 candidate coefficients from -15 to $+15$ (in steps of 0.03); the reported bounds are the smallest and largest candidate values not rejected at the 10% level, and are set to $\pm\infty$ when ± 15 is not rejected. Kleibergen-Paap F-statistics based on Conley standard errors are reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Correlation between the sum of shares and the instrument, migration, and income per capita

	Instrument		Average Immigrant Share, 1860-1920		Log income per capita in 2000	
	(1)	(2)	(3)	(4)	(5)	(6)
Expected instrument (μ_i)	0.954***	1.332***	4.481***	4.925***	11.001***	20.857***
	(0.014)	(0.071)	(1.167)	(2.136)	(2.364)	(7.898)
Observations	2935	2935	2935	2935	2935	2935
Baseline controls		✓		✓		✓

Notes: OLS regressions of the instrument, average migration share between 1860 and 1920 and log income per capita in 2000 on the sum of shares. Odd Columns do not include any controls; even Columns include all baseline controls of Table 4 of SNQ, including the *log(years of rail+1)* control. Conley standard errors using a five-degree window are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Alternative instruments: controlling for log(years of rail+1) and never-connected dummy

	Average in connected decades			Immigrant inflow in first-connected decade		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable:	Log income per capita, 2000					
Average Immigrant Share 1860-1920	8.532^{**}	8.172^{**}	48.806	4.540^{***}	4.179^{***}	-1.926
	(4.497)	(4.204)	(2719.172)	(1.773)	(1.686)	(4.254)
Observations	2935	2935	2935	2935	2935	2935
First-stage F-statistic	3.70	3.92	0.00	11.00	10.02	0.68
R^2 of IV on controls	0.659	0.789	0.789	0.960	0.759	0.978
Log(years of rail + 1)		✓	✓		✓	✓
Never connected	✓		✓	✓		✓

2SLS estimates based on the two alternative instruments: the average migrant share across all connected decades (Columns 1-3) and the migrant share in the first decade after connection (Columns 4-6). All models include the baseline controls of Table 4 of SNQ. Columns 2 and 5 additionally control for the *log(years of rail+1)* control and omit the binary indicator for never-connected counties; Columns 3 and 6 control for both. Conley standard errors using a five-degree window are shown in parentheses. Kleibergen-Paap F-statistics based on Conley standard errors and the R^2 of a regression of the instrument on all control variables are reported in the last rows.* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.